

Reading screening spirometry as a trajectory predicts incident airflow obstruction earlier

A landmark deep-learning analysis of 221,903 adults

Kyung Hyun Lee¹ · Dayoon Lee² · Jin Young Lee³ · Byung Jae Lee⁴

¹Digital Health, SAIHST, Sungkyunkwan University · ²Medical Device Management and Research, SAIHST
³Health Promotion Center, Samsung Medical Center · ⁴Division of Allergy, Samsung Medical Center, SKKU School of Medicine

No conflicts of interest to declare · **Contact: lkh256@gmail.com**



Normal airflow is not static: **the direction of change is an early warning, and the full sequence predicts best.**

6.4×
adjusted 5-year risk
falling vs rising

0.462 → 0.534
AUPRC
latest value → full sequence

+0.018 → +0.026
C-index advantage
0 → 2-year lead-time

ANALYSIS SETTING

Can a trajectory warn before the threshold is crossed?

Screening spirometry is often read one test at a time. A person can decline **inside the normal range** while every individual test still appears reassuring.

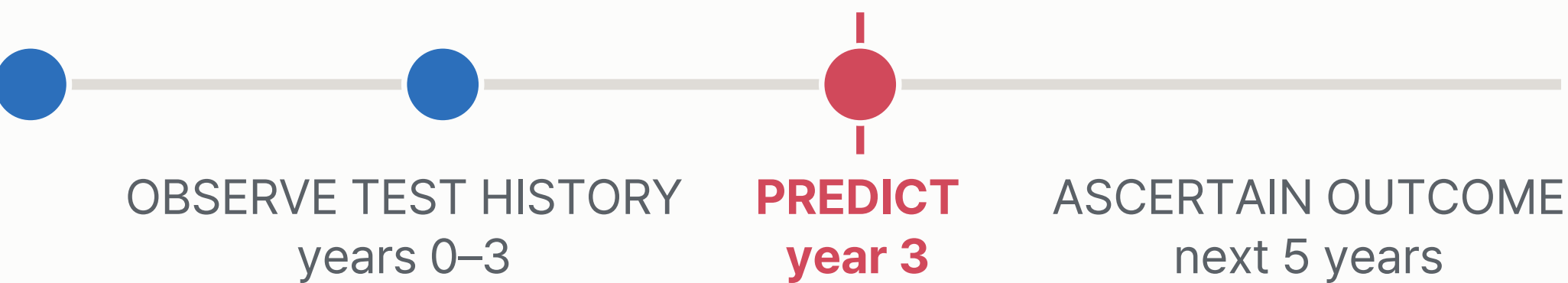
Does the trajectory add information beyond the latest value - and is a nonlinear model needed?

Who and when

221,903 adults · median age 48 [IQR 41–55]

557,503 health-screening spirometry tests

Entry: age ≥20 · first FEV1/FVC ≥75% · ≥2 tests before the fixed 3-year landmark.



Primary: one FEV1/FVC <0.70 reading.
Check: two consecutive low readings.

Three ways to read the same history

- Latest value** Last spirometry + age, sex, BMI (6 features)
- Linear trajectory** Latest value + slope, variability, test count, span
- Full sequence** ITE reads every time–variable–value token (PFT)

Patient-level 70/30 split · held-out n=12,823 · 452 events · pre-landmark predictors only.

CLINICAL SIGNAL

1. Direction matters before obstruction

Starting in the same normal band, a falling trajectory identified a much higher-risk group.

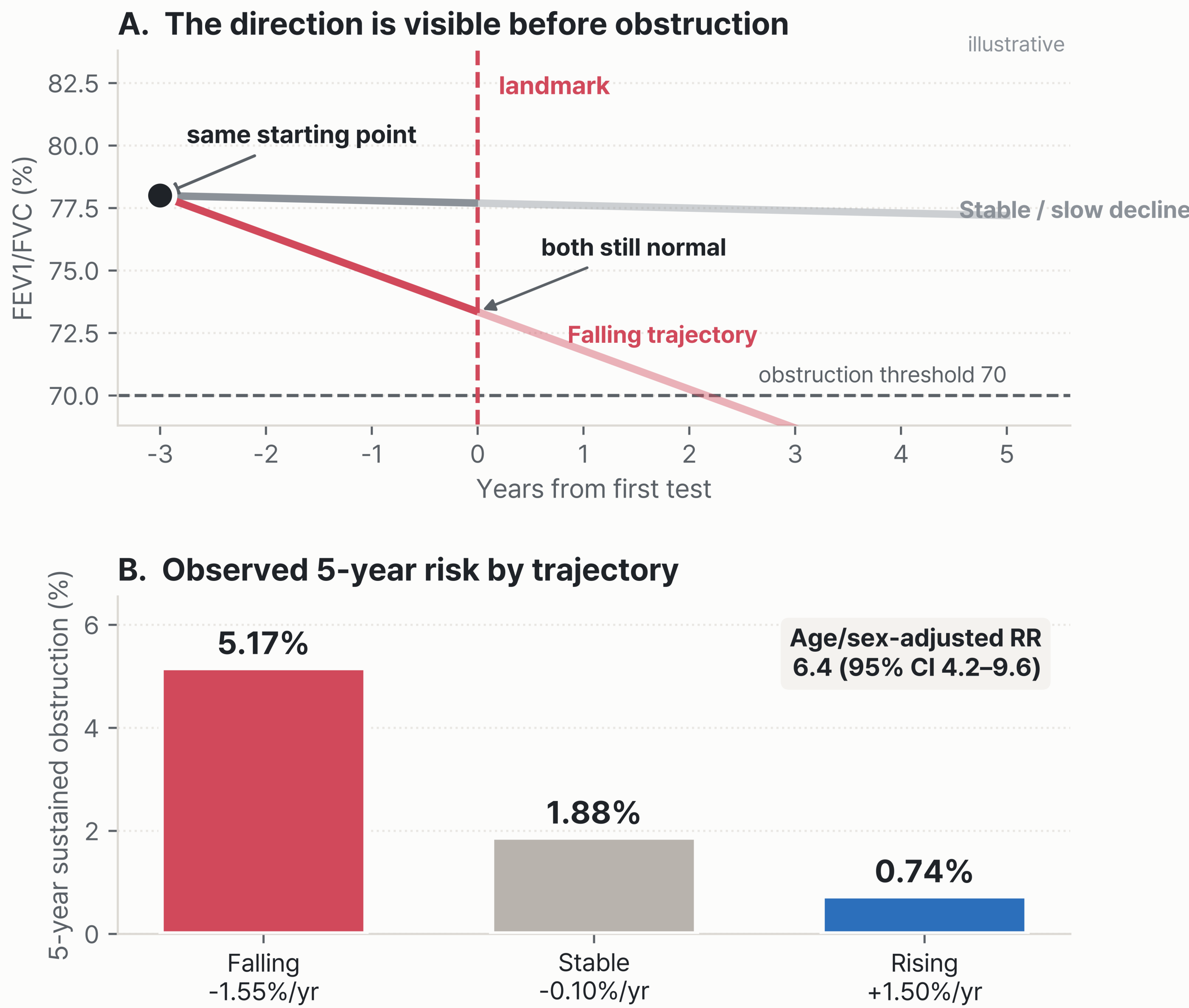


Figure 1. (A) Illustrative trajectories from a common normal baseline; the stable example declines slowly (–0.10%/year). (B) Observed 5-year sustained obstruction in the 75–80% band (two-reading outcome); events/n 181/3,502, 66/3,502 and 26/3,503.

At prediction, both trajectories are still above 0.70. The direction of arrival is what differs.

PREDICTIVE VALUE

2. More history improves prediction - further ahead

Both trajectory models improved the primary analysis; the full sequence model was best. Removing near-term events made its advantage larger.

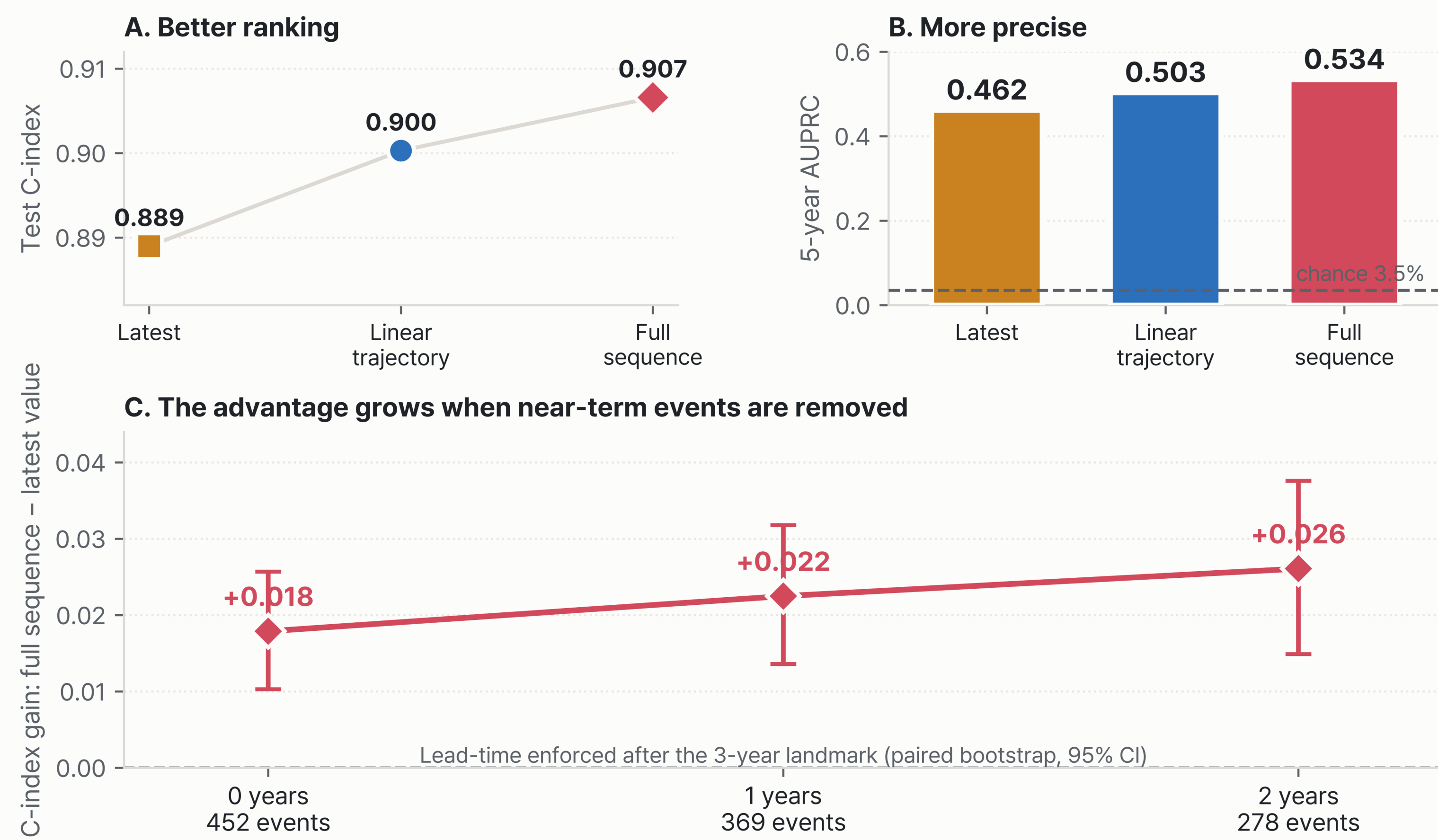


Figure 2. Held-out 3-year landmark performance (baseline FEV1/FVC ≥75%; n=12,823; 452 events). ITE values are 5-seed means; lead-time analyses exclude outcomes within 1–2 years after prediction.

ROBUSTNESS

Two-in-a-row outcome C-index 0.940 → 0.936 → 0.951
Baseline FEV1/FVC ≥80% C-index 0.825 → 0.824 → 0.843

WHAT THE RESULT MEANS

Trajectory information adds value beyond the latest measurement. The nonlinear gain persists under stricter definitions; the linear summary is not consistently better.

WHAT THIS SUGGESTS

Repeated screening spirometry should be read as a longitudinal signal, not as isolated tests. Change over time may flag risk years before a threshold-based label appears.

LIMITATION OF THIS STUDY

Retrospective single-centre health-screening data · no smoking, diagnosis or bronchodilator-response labels · spirometric airflow obstruction, not clinical COPD · internal held-out validation.

FUTURE WORKS

- Analyse clinically performed PFT records now being acquired from hospital departments.
- Conduct planned multicenter external validation of transportability, calibration and subgroup performance.